

Unequal-time density correlations in Gaussian adaptive circuits

Working derivation
(Dated: September 27, 2026)

We distinguish Gaussian states at individual times from a jointly Gaussian space-time process. We derive the usual Wick expression when the latter applies and an exact Gaussian-probe representation of ordered density correlations for an adaptive Markov circuit. Neither endpoint covariance nor the modular Hamiltonian alone determines correlations in physical circuit time.

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I. OBSERVABLE AND CONVENTION

Let $\hat{n}_i = \hat{c}_i^\dagger \hat{c}_i$ and $G_{ij}(s) = \text{Tr}[\hat{\rho}_s \hat{c}_i^\dagger \hat{c}_j]$. Here $i = (x, y, \mu)$ includes the orbital index, and G denotes the ordinary occupation correlator, not the centered covariance called $\mathbf{G}$ in saved files. For a fixed initial state at s and the subsequent trace-preserving circuit channel $\mathcal{E}_{t:s}$, define the ordered, connected correlator

$$K_{ij}(t, s) = \text{Tr}[\hat{n}_i \mathcal{E}_{t:s}(\hat{n}_j \hat{\rho}_s)], \quad (1)$$

$$C_{ij}(t, s) = K_{ij}(t, s) - \langle \hat{n}_i(t) \rangle \langle \hat{n}_j(s) \rangle. \quad (2)$$

The map is extended linearly to the generally non-Hermitian insertion $\hat{n}_j \hat{\rho}_s$. This is the circuit regression definition for fresh ancillas/Markov dynamics; any feedback memory must be retained or fixed by conditioning on the past record. It is not the correlation of two invasive projective density measurements. The real part is the symmetrized connected correlator. A cell-density correlation sums C_{ij} over both orbital indices.

II. JOINTLY GAUSSIAN DYNAMICS: WICK REDUCTION

For a number-conserving jointly Gaussian process, introduce

$$F_{ij}(t, s) = \langle \hat{c}_i^\dagger(t) \hat{c}_j(s) \rangle, \quad H_{ij}(t, s) = \langle \hat{c}_i(t) \hat{c}_j^\dagger(s) \rangle. \quad (3)$$

The anomalous contractions vanish. Wick's theorem gives

$$C_{ij}(t, s) = F_{ij}(t, s) H_{ij}(t, s). \quad (4)$$

This is an elementwise product, not a matrix product or generally a modulus squared. In particular, $C_{ij}(s, s) = G_{ij}(\delta_{ij} - G_{ji})$: the diagonal equals $G_{ii}(1 - G_{ii})$, whereas distinct-mode correlations equal $-|G_{ij}|^2$.

For quadratic unitary evolution write $\hat{\mathbf{c}}(t) = U(t, s) \hat{\mathbf{c}}(s)$. Direct substitution gives

$$C_{ij}(t, s) = [U^* G(s)]_{ij} [U(\mathbf{1} - G(s)^T)]_{ij}. \quad (5)$$

The same expression with U replaced by X applies to a number-conserving Gaussian dilation $\hat{\mathbf{c}}(t) = X \hat{\mathbf{c}}(s) + \hat{\boldsymbol{\eta}}$ with independent, zero-mean Gaussian incoming noise and a Gaussian joint input. Dynamic correlations in quadratic open fermionic systems with linear Lindblad operators provide a standard setting for this construction [1].

Gaussian conditional states alone do not guarantee these assumptions after averaging a measurement record. For example, a single mode with occupation p subjected to an equally weighted identity/parity unitary has vanishing averaged one-body propagator. Nevertheless its density is conserved and $C_{nn} = p(1 - p)$. Multiplying the separately averaged one-body contractions would incorrectly give zero. Gaussianity also does not fix a power-law exponent: the same Gaussian initial state can be evolved under different dynamics.

III. EXACT GAUSSIAN PROBES FOR THE ADAPTIVE CIRCUIT

The following identities do not assume the nonselective channel is Gaussian. Define two perturbed initial states at s :

$$\hat{\rho}_\epsilon^f = \frac{e^{\epsilon \hat{n}_j/2} \hat{\rho}_s e^{\epsilon \hat{n}_j/2}}{\text{Tr}(e^{\epsilon \hat{n}_j} \hat{\rho}_s)}, \quad (6)$$

$$\hat{\rho}_\epsilon^p = e^{-i\epsilon \hat{n}_j} \hat{\rho}_s e^{i\epsilon \hat{n}_j}, \quad f_i^a(\epsilon) = \text{Tr}[\hat{n}_i \mathcal{E}_{t:s}(\hat{\rho}_\epsilon^a)]. \quad (7)$$

Both inputs remain Gaussian when $\hat{\rho}_s$ is Gaussian. Differentiating the normalized filter and the unitary phase rotation yields

$$\boxed{\text{Re } C_{ij} = \partial_\epsilon f_i^f|_0, \quad \text{Im } C_{ij} = \frac{1}{2} \partial_\epsilon f_i^p|_0.} \quad (8)$$

Indeed the two state derivatives are $\{\hat{n}_j, \hat{\rho}_s\}/2 - G_{jj}\hat{\rho}_s$ and $-i[\hat{n}_j, \hat{\rho}_s]$, respectively. Hermiticity preservation relates the two operator orderings by complex conjugation.

These probes have explicit single-particle formulas. With $P_j = |j\rangle\langle j|$, $u = e^\epsilon - 1$, $p = G_{jj}$, and $D = e^{\epsilon P_j/2}$,

$$G_\epsilon^f = D \left[G - \frac{u}{1 + up} G P_j G \right] D, \quad (9)$$

$$G_\epsilon^p = e^{i\epsilon P_j} G e^{-i\epsilon P_j}, \quad (10)$$

$$\partial_\epsilon G_\epsilon^f|_0 = \frac{1}{2} (P_j G + G P_j) - G P_j G. \quad (11)$$

The first identity follows by Wick contraction and $e^{\epsilon \hat{n}_j} = \hat{1} + u \hat{n}_j$; it also holds for pure Gaussian states by continuity.

For each probe, evolve the *same circuit rule* with its own correct Born probabilities, not a frozen sequence of outcomes without likelihood weights. The difference quotient at $\pm\epsilon$ approximates the derivatives above; finite- ϵ convergence and Monte Carlo uncertainty must be tested. Common random numbers can couple runs, but outcome probabilities must still be recomputed. At $s > 0$, one may condition on a Gaussian past trajectory. To obtain the full ensemble connected correlator, average the unconnected correlators first and subtract ensemble means: averaging conditional connected correlators alone misses the covariance of conditional means.

IV. IMPLICATIONS FOR THE PRESENT DATA

The stored spectra, entropy contours, and final covariance determine equal-time observables but not the two-time propagator or the operator-insertion channel. The physical-time propagator is not $e^{-i h_{\text{modular}}(t-s)}$ unless that evolution is explicitly chosen. New probe evolution or a faithful recorded instrument with insertion propagation is required for the circuit correlator. The scalar entropy histories can instead supply classical trajectory-level two-time statistics; those are different observables and are nonstationary during purification. Power-law-looking purification is not by itself evidence of a stationary power-law quantum correlator.

a. Validation. `test.unequal_time_gaussian.py` checks the Wick expression, both single-particle probe updates, and the complete complex density correlator against exact three-mode Fock-space calculations for quadratic unitary evolution and a mixture of Gaussian unitaries. It also checks the dephasing counterexample.

[1] P. Kos and T. Prosen, *Time-dependent Correlation Functions in Open Quadratic Fermionic Systems*, (2017), [doi:10.1088/1742-5468/aa9681](https://doi.org/10.1088/1742-5468/aa9681); [arXiv:1708.06919](https://arxiv.org/abs/1708.06919).